

Name: _____



Problem solving with sequences

Task A: Generalising from non-sequential terms

1) Fill in the missing terms in these arithmetic sequences.

Term number (n)	1	2	3	4	5	6
a)				1	10	
b)		1			10	
c)		1		10		
d)	10				1	

2) What happens when you try to find the missing terms in this sequence? What does that tell you about this sequence?

2, , 32, , 98, ...

3) The 23rd term of an arithmetic sequence is 187 and the 45th term is 363.

a) Find the 100th term.

b) Will 3634 be a term in this sequence?



Task B: Fractional sequences

1) Jun and Laura are looking at a sequence.

$$\frac{1}{5}, \frac{1}{10}, \frac{1}{15}, \frac{1}{20}, \frac{1}{25}, \dots$$



5, 10, 15, 20, 25, ...

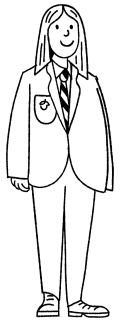
This is an increasing sequence.

a) Justify why Jun is wrong.

5, 10, 15, 20, 25, ... That's $5n$.

This is an arithmetic sequence.

b) Justify why Laura is wrong.



2) Write an n^{th} term for these fractional sequences.

a) $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \frac{1}{6}, \dots$

b) $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6}, \dots$

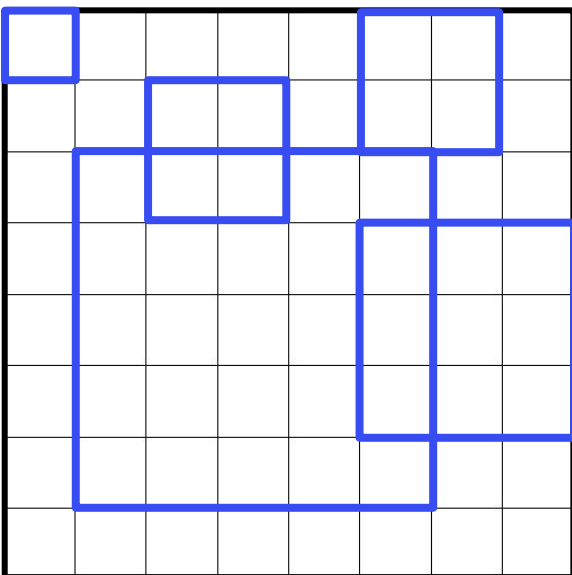
c) $\frac{1}{3}, \frac{2}{5}, \frac{3}{7}, \frac{4}{9}, \frac{5}{11}, \dots$

d) $\frac{1}{3}, \frac{6}{5}, \frac{11}{7}, \frac{16}{9}, \frac{21}{11}, \dots$

e) $\frac{1}{1}, \frac{1}{4}, \frac{1}{9}, \frac{1}{16}, \frac{1}{25}, \dots$

Task C: Pattern spotting

1) How many squares are there on a chess board?



Don't forget to consider all the squares!